

Question 1:

Find $\lim_{x \rightarrow 3} \frac{x^2 - 12x + 27}{x - 3}$

a) Apply L'Hopital's Rule and find derivatives: $\frac{D_x(x^2 - 12x + 27)}{D_x(x - 3)} = ?$

b) Answer: $\lim_{x \rightarrow 3} \frac{x^2 - 12x + 27}{x - 3} = \underline{\hspace{4cm}}$

Question 2:

$$\text{Find } \lim_{x \rightarrow 0} \frac{\sqrt{4-x^2} - 2}{x} = \lim_{x \rightarrow 0} \frac{(\sqrt{4-x^2} - 2)}{(x)} \cdot \frac{(\sqrt{4-x^2} + 2)}{(\sqrt{4-x^2} + 2)},$$

$$\text{Note: } (A-B)(A+B) = A^2 - B^2$$

Note: L'Hopital's Rule is not useful for this problem.

$$\text{a) Simplify: } \frac{(\sqrt{4-x^2} - 2)}{(x)} \cdot \frac{(\sqrt{4-x^2} + 2)}{(\sqrt{4-x^2} + 2)} = ?$$

$$\text{b) Answer: } \lim_{x \rightarrow 0} \frac{\sqrt{4-x^2} - 2}{x} = \underline{\hspace{2cm}}$$

Question 3:

Find $\lim_{x \rightarrow 0^+} \frac{e^x - (1 + x)}{x^2}$

a) Apply L'Hopital's Rule and find derivatives: $\frac{D_x(e^x - (1 + x))}{D_x(x^2)} = ?$

Note: Need to apply L'Hopital's Rule several times.

b) Answer: $\lim_{x \rightarrow 0^+} \frac{e^x - (1 + x)}{x^2} = \underline{\hspace{2cm}}$

Question 4:

Find $\lim_{x \rightarrow 2} \frac{x^6 - 64}{x^4 - 16}$

a) Apply L'Hopital's Rule and find derivatives: $\frac{x^6 - 64}{x^4 - 16} = ?$

Note: Need to apply L'Hopital's Rule several times.

b) Answer: $\lim_{x \rightarrow 2} \frac{x^6 - 64}{x^4 - 16} = \underline{\hspace{2cm}}$

Question 5:

Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 7x}$

a) Apply L'Hopital's Rule and find derivatives: $\frac{D_x(\sin 2x)}{D_x(\sin 7x)} = ?$

b) Answer: $\lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 7x} =$ _____

Question 6:

Find $\lim_{x \rightarrow \infty} \frac{x^2}{e^{x/4}}$

a) Apply L'Hopital's Rule and find derivatives: $\frac{D_x(x^2)}{D_x(e^{x/4})} = ?$

Note: Need to apply L'Hopital's Rule several times.

b) Answer: $\lim_{x \rightarrow \infty} \frac{x^2}{e^{x/4}} =$ _____

Question 7:

$$\text{Find } \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}} = \lim_{x \rightarrow \infty} \frac{x / \sqrt{x^2}}{\sqrt{x^2 + 1} / \sqrt{x^2}} = \lim_{x \rightarrow \infty} \frac{x / x}{\sqrt{\frac{x^2 + 1}{x^2}}}$$

$$\text{Note: } \sqrt{x^2} = \begin{cases} x & \text{if } x \rightarrow \infty \\ -x & \text{if } x \rightarrow -\infty \end{cases}$$

Hint: L'Hôpital's Rule is not useful for this problem.

a) Simplify: $\frac{x / x}{\sqrt{\frac{x^2 + 1}{x^2}}} = ?$

b) $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}} =$ _____

Question 8:

Find $\lim_{x \rightarrow 0^+} x^{5/x}$

Hint: Let $y = \lim_{x \rightarrow 0^+} x^{5/x} \Rightarrow \ln y = \ln \left(\lim_{x \rightarrow 0^+} x^{5/x} \right) \Rightarrow \ln \left(\lim_{x \rightarrow 0^+} x^{5/x} \right) = \lim_{x \rightarrow 0^+} [\ln x^{5/x}]$

Hint: $(\infty)(\infty) = \infty$; $(-\infty)(\infty) = -\infty$; $(\infty) + (\infty) = \infty$; $(-\infty) + (-\infty) = -\infty$;

$(\infty) - (\infty) = \text{indeterminate}$; $\frac{\infty}{\infty} = \text{indeterminate}$

a) As $x \rightarrow 0^+$, $\frac{5}{x} \rightarrow \underline{\hspace{2cm}}?$

b) As $x \rightarrow 0^+$, $\ln x \rightarrow \underline{\hspace{2cm}}?$

c) $y = \lim_{x \rightarrow 0^+} x^{5/x} = \underline{\hspace{2cm}}?$

Question 9:

Find $\lim_{x \rightarrow \infty} x^{1/x}$

Hint: Let $y = \lim_{x \rightarrow \infty} x^{1/x} \Rightarrow \ln y = \ln \left(\lim_{x \rightarrow \infty} x^{1/x} \right) \Rightarrow \ln \left(\lim_{x \rightarrow \infty} x^{1/x} \right) = \lim_{x \rightarrow \infty} [\ln x^{1/x}]$

Hint: $(\infty)(\infty) = \infty$; $(-\infty)(\infty) = -\infty$; $(\infty) + (\infty) = \infty$; $(-\infty) + (-\infty) = -\infty$;

$(\infty) - (\infty) = \text{indeterminate}$; $\frac{\infty}{\infty} = \text{indeterminate}$

a) As $x \rightarrow \infty$, $\frac{1}{x} \rightarrow \underline{\hspace{2cm}} ?$

b) As $x \rightarrow \infty$, $\ln x \rightarrow \underline{\hspace{2cm}} ?$

c) $y = \lim_{x \rightarrow \infty} x^{1/x} = \underline{\hspace{2cm}} ?$

Question 10:

Find $\lim_{x \rightarrow 0^+} (x + 5)^{1/x}$

Hint: Let $y = \lim_{x \rightarrow 0^+} (x + 5)^{1/x} \Rightarrow \ln y = \ln \left(\lim_{x \rightarrow 0^+} (x + 5)^{1/x} \right) \Rightarrow \ln \left(\lim_{x \rightarrow 0^+} (x + 5)^{1/x} \right) = \lim_{x \rightarrow 0^+} \left[\ln (x + 5)^{1/x} \right]$

Hint: $(\infty)(\infty) = \infty$; $(-\infty)(\infty) = -\infty$; $(\infty) + (\infty) = \infty$; $(-\infty) + (-\infty) = -\infty$;

$(\infty) - (\infty) = \text{indeterminate}$; $\frac{\infty}{\infty} = \text{indeterminate}$

a) As $x \rightarrow 0^+$, $\frac{1}{x} \rightarrow \underline{\quad ? \quad}$

b) As $x \rightarrow 0^+$, $\ln(x + 5) \rightarrow \underline{\quad ? \quad}$

c) $y = \lim_{x \rightarrow 0^+} (x + 5)^{1/x} = \underline{\quad ? \quad}$

